Conclusion

1 Architectural contribution

A 0.8B-parameter CPU classifier replaces the LLM consultant. This eliminates the 21% JSON-schema fallback rate at scale, lifts state accuracy by 2.14× (+29.45 pp), and raises the unified score by +19.25 points over the GPT-4o + SocratTeachLLM paper baseline. All at $0 API cost on one 32 GB consumer GPU.

2 Methodological contribution

Five converging diagnostics establish that ROUGE and BLEU on this benchmark systematically reward training-data memorization rather than transferable teaching capability. The five diagnostics are surface-form rank inversion, monotonic n-gram-length scaling, cross-lingual translation reproduction, clean-probe collapse, and within-architecture base-model ablation.

3 Limitations and future work

Result is specific to this teacher, prompt, and consultant (Chinese-language SocratDataset). Stage 2b QLoRA adapter on the Gemma teacher trained successfully, but evaluation is blocked by a train/serve prompt-format mismatch. Native English-domain benchmark and bilingual classifier co-training are next.

The same five-diagnostic audit applies to any Socratic-teaching benchmark.